# 偏微分方程第六次作业

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**1.**

(a) 设 $\boldsymbol{E}$ 和 $\boldsymbol{B}$ 满足 Maxwell 方程组

$$\begin{cases} \boldsymbol{E}_t = \operatorname{curl} \boldsymbol{B}, \\ \boldsymbol{B}_t = -\operatorname{curl} \boldsymbol{E}, \\ \operatorname{div} \boldsymbol{B} = \operatorname{div} \boldsymbol{E} = 0. \end{cases}$$

求证: $\boldsymbol{E}_{tt} - \Delta \boldsymbol{E} = \boldsymbol{0}$, $\boldsymbol{B}_{tt} - \Delta \boldsymbol{B} = \boldsymbol{0}$;

(b) 设 $\boldsymbol{u}$ 满足线性弹性力学的演化方程

$$\boldsymbol{u}_{tt} - \mu \Delta \boldsymbol{u} - (\lambda + \mu) \nabla (\operatorname{div} \boldsymbol{u}) = 0, \quad (t, \boldsymbol{x}) \in \mathbb{R}^+ \times \mathbb{R}^3,$$

其中 $\lambda, \mu$ 为正常数. 求证: $v := \operatorname{div} \boldsymbol{u}$ 和 $\boldsymbol{w} := \operatorname{curl} \boldsymbol{u}$ 各自满足波方程, 但传播速度不同.

**解答:**

(a) 注意到

$$\boldsymbol{E}_{tt} = \operatorname{curl} \boldsymbol{B}_t = -\operatorname{curl}(\operatorname{curl} \boldsymbol{E}) = -(\nabla(\operatorname{div} \boldsymbol{E}) - \Delta \boldsymbol{E}) = \Delta \boldsymbol{E},$$

故 $\boldsymbol{E}_{tt} - \Delta \boldsymbol{E} = \boldsymbol{0}$. 同理可得 $\boldsymbol{B}_{tt} - \Delta \boldsymbol{B} = \boldsymbol{0}$.

(b) 注意到

$$v_{tt} = \mu \operatorname{div}(\Delta \boldsymbol{u}) + (\lambda + \mu) \operatorname{div}(\nabla(\operatorname{div} \boldsymbol{u})) = \mu \Delta(\operatorname{div} \boldsymbol{u}) + (\lambda + \mu) \Delta(\operatorname{div} \boldsymbol{u}) = (\lambda + 2\mu) \Delta v,$$

故 $v_{tt} - (\lambda + 2\mu) \Delta v = 0$; 以及

$$\boldsymbol{w}_{tt} = \mu \operatorname{curl}(\Delta \boldsymbol{u}) + (\lambda + \mu) \operatorname{curl}(\nabla(\operatorname{div} \boldsymbol{u})) = \mu \Delta(\operatorname{curl} \boldsymbol{u}) = \mu \Delta \boldsymbol{w},$$

从而 $\boldsymbol{w}_{tt} - \mu \Delta \boldsymbol{w} = 0$, 并且二者的传播速度不同.

□

**2.** 设 $a$ 为正常数. 求证: 三维波方程

$$u_{tt} - a^2\Delta u = 0, \quad (t, \boldsymbol{x}) \in \mathbb{R}_+ \times \mathbb{R}^3$$

的所有径向对称的解都可以表示为

$$u(t, \boldsymbol{x}) = \frac{F(|\boldsymbol{x}| - at) + G(|\boldsymbol{x}| + at)}{|\boldsymbol{x}|}, \quad \boldsymbol{x} \neq \boldsymbol{0}.$$

**解答:** 设 $u = u(t, r)$, 注意到

$$u_{x_k} = u_r \frac{x_k}{r}, \quad u_{x_k x_k} = u_{rr}\left(\frac{x_k}{r}\right)^2 + u_r \frac{r^2 - x_k^2}{r^3} \implies \Delta u = u_{rr} + \frac{2}{r}u_r.$$

故

$$u_{tt} - a^2\left(u_{rr} + \frac{2}{r}u_r\right) = 0, \quad (t, r) \in \mathbb{R}_+ \times \mathbb{R}_+.$$

$\forall\, r > 0$, 令 $v = ru$, 即 $u = v/r$, 代入上述方程化简可得 $v_{tt} - a^2 v_{rr} = 0$. 再注意到一维波方程的解 $v$ 可以表示为 $v(t, r) = F(r - at) + G(r + at)$, 从而

$$u(t, \boldsymbol{x}) = \frac{F(|\boldsymbol{x}| - at) + G(|\boldsymbol{x}| + at)}{|\boldsymbol{x}|}, \quad \boldsymbol{x} \neq \boldsymbol{0}.$$

□

**3.** 设 $a$ 为正常数. 求解初值问题

$$\begin{cases} u_{tt} - a^2\Delta u = 0, & (t, \boldsymbol{x}) \in \mathbb{R}_+ \times \mathbb{R}^3, \\ u(0, \boldsymbol{x}) = x^2 + y^2 z, u_t(0, \boldsymbol{x}) = y + 1, & \boldsymbol{x} = (x, y, z) \in \mathbb{R}^3. \end{cases}$$

**解答:** 直接计算可得

$$\begin{aligned}
&\oint_{\partial B((x_0,y_0,z_0),at)} (x^2 + y^2 z)\,\mathrm{d}S \\
&= \frac{1}{4\pi(at)^2}\int_0^{2\pi}\int_0^{\pi} (at)^2 \sin\varphi((x_0 + at\sin\varphi\cos\theta)^2 + (y_0 + at\sin\varphi\sin\theta)^2(z_0 + at\cos\varphi))\,\mathrm{d}\varphi\,\mathrm{d}\theta \\
&= \frac{1}{4\pi}\int_0^{\pi} \sin\varphi((2\pi x_0^2 + \pi(at)^2\sin^2\varphi) + (2\pi y_0^2 + \pi(at)^2\sin^2\varphi)(z_0 + at\cos\varphi))\,\mathrm{d}\varphi \\
&= \frac{1}{4}\int_0^{\pi} \sin\varphi((2x_0^2 + (at)^2\sin^2\varphi) + (2y_0^2 + (at)^2\sin^2\varphi)z_0)\,\mathrm{d}\varphi \\
&= \frac{1}{4}\int_0^{\pi} \left((2x_0^2 + 2y_0^2 z_0)\sin\varphi + (at)^2(z_0 + 1)\sin^3\varphi\right)\mathrm{d}\varphi
\end{aligned}$$

$$= \frac{1}{4}\left(2(2x_0^2 + 2y_0^2 z_0) + \frac{4}{3}(at)^2(z_0+1)\right)$$
$$= x_0^2 + y_0^2 z_0 + \frac{1}{3}(at)^2(z_0+1),$$

故

$$\partial_t\left(t \oint_{\partial B((x_0,y_0,z_0),at)} (x^2 + y^2 z)\,\mathrm{d}S\right) = x_0^2 + y_0^2 z_0 + a^2 t^2 (z_0+1).$$

同理可得

$$\oint_{\partial B((x_0,y_0,z_0),at)} (y+1)\,\mathrm{d}S = \frac{1}{4\pi(at)^2}\int_0^{2\pi}\int_0^{\pi} (at)^2 \sin\varphi (y_0 + at\sin\varphi\sin\theta + 1)\,\mathrm{d}\varphi\,\mathrm{d}\theta$$
$$= \frac{1}{4\pi}\int_0^{\pi} 2\pi(y_0+1)\sin\varphi\,\mathrm{d}\varphi = y_0 + 1,$$

从而

$$t \oint_{\partial B((x_0,y_0,z_0),at)} (y+1)\,\mathrm{d}S = t(y_0+1).$$

因此

$$u(t,x,y,z) = x^2 + y^2 z + a^2 t^2 (z+1) + t(y+1).$$

□

**4.** 设 $a$ 为正常数, 函数 $\varphi(x), \psi(x) \in C^2[0,+\infty), g(t) \in C^2[0,+\infty)$ 满足相容性条件

$$g(0) = \varphi(0), \quad g'(0) = \psi(0), \quad g''(0) = a^2\varphi''(0).$$

求解半空间问题

$$\begin{cases} u_{tt} - a^2 u_{xx} = 0, & (t,x) \in \mathbb{R}_+ \times \mathbb{R}_+, \\ u(0,x) = \varphi(x), u_t(0,x) = \psi(x), & x \in \mathbb{R}_{\geqslant 0}, \\ u(t,0) = g(t), & t \in \mathbb{R}_{\geqslant 0}. \end{cases}$$

**解答:** 令 $v(t,x) := u(t,x) - g(t)$, 则 $v$ 满足方程

$$\begin{cases} v_{tt} - a^2 v_{xx} = -g''(t) := F(t,x), & (t,x) \in \mathbb{R}_+ \times \mathbb{R}_+, \\ v(0,x) = \varphi(x) - g(0) := G(x), & x \in \mathbb{R}_{\geqslant 0}, \\ v_t(0,x) = \psi(x) - g'(0) := H(x), & x \in \mathbb{R}_{\geqslant 0}, \\ v(t,0) = 0, & t \in \mathbb{R}_{\geqslant 0}. \end{cases}$$

将 $F(t,x), G(x), H(x)$ 分别对 $x$ 做奇延拓得到 $\widetilde{F}(t,x), \widetilde{G}(x), \widetilde{H}(x)$, 故

$$\begin{aligned} v(t,x) &= \frac{1}{2a}\int_{x-at}^{x+at} \widetilde{H}(\lambda)\,\mathrm{d}\lambda + \frac{1}{2}\left(\widetilde{G}(x+at)+\widetilde{G}(x-at)\right) + \frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)} \widetilde{F}(\tau,\lambda)\,\mathrm{d}\lambda\,\mathrm{d}\tau \\ &= \frac{1}{2a}\int_{|x-at|}^{x+at} H(\lambda)\,\mathrm{d}\lambda + \frac{1}{2}\left(G(x+at)+\operatorname{sign}(x-at)G(|x-at|)\right) + \frac{1}{2a}\int_0^t\int_{|x-a(t-\tau)|}^{x+a(t-\tau)} F(\tau,\lambda)\,\mathrm{d}\lambda\,\mathrm{d}\tau, \end{aligned}$$

从而

$$\begin{aligned} &u(t,x) \\ &= v(t,x) + g(t) \\ &= \frac{1}{2a}\int_{|x-at|}^{x+at} H(\lambda)\,\mathrm{d}\lambda + \frac{1}{2}\left(G(x+at)+\operatorname{sign}(x-at)G(|x-at|)\right) + \frac{1}{2a}\int_0^t\int_{|x-a(t-\tau)|}^{x+a(t-\tau)} F(\tau,\lambda)\,\mathrm{d}\lambda\,\mathrm{d}\tau + g(t). \end{aligned}$$

□

**5.** 设函数 $u \in C^2([0,+\infty)\times\mathbb{R}^3)$ 满足初值问题

$$\begin{cases} u_{tt} - a^2\Delta u = 0, & (t,\boldsymbol{x}) \in \mathbb{R}_+\times\mathbb{R}^3, \\ u(0,\boldsymbol{x}) = \varphi(\boldsymbol{x}), u_t(0,\boldsymbol{x}) = \psi(\boldsymbol{x}) & \boldsymbol{x}\in\mathbb{R}^3, \end{cases}$$

其中 $a$ 为正常数, $\varphi,\psi \in C_0^\infty(\mathbb{R}^3)$. 求证: 存在常数 $C$, 使得 $\forall\,(t,\boldsymbol{x})\in\mathbb{R}_+\times\mathbb{R}^3$, 有 $|u(t,\boldsymbol{x})| \leqslant C/t$.

**解答:** 取固定的 $R>0$ 使得 $\operatorname{supp}\varphi, \operatorname{supp}\psi \subseteq B(\mathbf{0},R)$. 注意到 $\forall\,(t,\boldsymbol{x})\in\mathbb{R}_+\times\mathbb{R}^3$, 有

$$u(t,\boldsymbol{x}) = \fint_{\partial B(\boldsymbol{x},at)} \left(\varphi(\boldsymbol{y}) + \nabla\varphi(\boldsymbol{y})\cdot(\boldsymbol{y}-\boldsymbol{x}) + t\psi(\boldsymbol{y})\right)\mathrm{d}S(\boldsymbol{y}),$$

故

$$\begin{aligned} |u(t,\boldsymbol{x})| &\leqslant \frac{1}{4\pi(at)^2}\left(\|\varphi\|_{L^\infty} + at\,\|\nabla\varphi\|_{L^\infty} + t\,\|\psi\|_{L^\infty}\right)\int_{\partial B(\boldsymbol{x},at)} \mathbf{1}_{\boldsymbol{y}\in B(\mathbf{0},R)}\,\mathrm{d}S(\boldsymbol{y}) \\ &\lesssim \frac{1}{t^2}(1+t)\min\{4\pi(at)^2, 4\pi R^2\} \\ &\lesssim \frac{1}{t^2}(1+t)\min\{t^2,1\} \lesssim \frac{1}{t}, \end{aligned}$$

这里用到了若交集非空, 任取其中一点 $\boldsymbol{y}_0$, 则交集包含在 $\partial B(\boldsymbol{x},at)\cap B(\boldsymbol{y}_0,2R)$ 中, 再由球冠面积公式, 该曲面面积至多为 $4\pi R^2$. □

**6.** 设 $g\in C^3(\mathbb{R}^3), h\in C^2(\mathbb{R}^3), f\in C^2([0,+\infty)\times\mathbb{R}^3)$, 定义

$$\begin{aligned}u(t,\boldsymbol{x}):=&\frac{1}{4\pi t^2}\iint_{\partial B(\boldsymbol{x},t)}[g(\boldsymbol{y})+\nabla g(\boldsymbol{y})\cdot(\boldsymbol{y}-\boldsymbol{x})+th(\boldsymbol{y})]\,\mathrm{d}S(\boldsymbol{y})+\\&+\frac{1}{4\pi}\int_0^t\frac{1}{t-\tau}\iint_{\partial B(\boldsymbol{x},t-\tau)}f(\tau,\boldsymbol{y})\,\mathrm{d}S(\boldsymbol{y})\,\mathrm{d}\tau.\end{aligned}$$

求证:

(a) $u\in C^2([0,+\infty)\times\mathbb{R}^3)$;

(b) $u_{tt}-\Delta u=f, \forall\,(t,\boldsymbol{x})\in\mathbb{R}_+\times\mathbb{R}^3$;

(c) $\forall\,\boldsymbol{x}_0\in\mathbb{R}^3$, 有

$$\lim_{(t,\boldsymbol{x})\to(0,\boldsymbol{x}_0)}u(t,\boldsymbol{x})=g(\boldsymbol{x}_0),\qquad\lim_{(t,\boldsymbol{x})\to(0,\boldsymbol{x}_0)}u_t(t,\boldsymbol{x})=h(\boldsymbol{x}_0).$$

**解答:**

(a) 注意到

$$\begin{aligned}u(t,\boldsymbol{x}):=&\frac{1}{4\pi}\iint_{\partial B(\boldsymbol{0},1)}[g(\boldsymbol{x}+t\boldsymbol{z})+\nabla g(\boldsymbol{x}+t\boldsymbol{z})\cdot t\boldsymbol{z}+th(\boldsymbol{x}+t\boldsymbol{z})]\,\mathrm{d}S(\boldsymbol{z})+\\&+\frac{1}{4\pi}\int_0^t(t-\tau)\iint_{\partial B(\boldsymbol{0},1)}f(\tau,\boldsymbol{x}+(t-\tau)\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})\,\mathrm{d}\tau,\end{aligned}$$

故 $u\in C^2([0,+\infty)\times\mathbb{R}^3)$.

(b) 忽略系数 $1/4\pi$, 只需验证

$$u_2(t,\boldsymbol{x})=t\iint_{\partial B(\boldsymbol{0},1)}h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})$$

满足 $\partial_t^2u_2-\Delta u_2=0$. 事实上,

$$\Delta u_2(t,\boldsymbol{x})=t\iint_{\partial B(\boldsymbol{0},1)}\Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z}),$$

同时

$$\begin{aligned}\partial_tu_2(t,\boldsymbol{x})&=\iint_{\partial B(\boldsymbol{0},1)}h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})+t\iint_{\partial B(\boldsymbol{0},1)}\boldsymbol{z}\cdot\nabla h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})\\&=\iint_{\partial B(\boldsymbol{0},1)}h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})+t^2\iiint_{B(\boldsymbol{0},1)}\Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}\boldsymbol{z}\\&=\iint_{\partial B(\boldsymbol{0},1)}h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})+\frac{1}{t}\iiint_{B(\boldsymbol{x},t)}\Delta h(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y},\\\partial_t^2u_2(t,\boldsymbol{x})&=t\iiint_{B(\boldsymbol{0},1)}\Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}\boldsymbol{z}-\frac{1}{t^2}\iiint_{B(\boldsymbol{x},t)}\Delta h(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}+\frac{1}{t}\iint_{\partial B(\boldsymbol{x},t)}\Delta h(\boldsymbol{y})\,\mathrm{d}S(\boldsymbol{y})\end{aligned}$$

$$
\begin{aligned}
&= t \iiint_{B(\mathbf{0},1)} \Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}\boldsymbol{z} - t\iiint_{B(\mathbf{0},1)} \Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}\boldsymbol{z} + t\iint_{\partial B(\mathbf{0},1)} \Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z}) \\
&= t\iint_{\partial B(\mathbf{0},1)} \Delta h(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z}),
\end{aligned}
$$

因此 $\partial_t^2 u_2 - \Delta u_2 = 0$.

(c) 注意到

$$
\begin{aligned}
u(t,\boldsymbol{x}) :=& \frac{1}{4\pi}\iint_{\partial B(\mathbf{0},1)} [g(\boldsymbol{x}+t\boldsymbol{z}) + \nabla g(\boldsymbol{x}+t\boldsymbol{z})\cdot t\boldsymbol{z} + th(\boldsymbol{x}+t\boldsymbol{z})]\,\mathrm{d}S(\boldsymbol{z}) + \\
&+ \frac{1}{4\pi}\int_0^t (t-\tau)\iint_{\partial B(\mathbf{0},1)} f(\tau,\boldsymbol{x}+(t-\tau)\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})\,\mathrm{d}\tau,
\end{aligned}
$$

故显然有 $\lim\limits_{(t,\boldsymbol{x})\to(0,\boldsymbol{x}_0)} u(t,\boldsymbol{x}) = g(\boldsymbol{x}_0)$. 再利用

$$
\frac{1}{4\pi}\iint_{\partial B(\mathbf{0},1)} [g(\boldsymbol{x}+t\boldsymbol{z}) + \nabla g(\boldsymbol{x}+t\boldsymbol{z})\cdot t\boldsymbol{z}]\,\mathrm{d}S(\boldsymbol{z}) = \frac{1}{4\pi}\partial_t\left(t\iint_{\partial B(\mathbf{0},1)} g(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})\right),
$$

以及由 (b) 知

$$
\partial_t^2\left(t\iint_{\partial B(\mathbf{0},1)} g(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z})\right) = t\iint_{\partial B(\mathbf{0},1)} \Delta g(\boldsymbol{x}+t\boldsymbol{z})\,\mathrm{d}S(\boldsymbol{z}),
$$

从而

$$
\lim_{(t,\boldsymbol{x})\to(0,\boldsymbol{x}_0)} u_t(t,\boldsymbol{x}) = h(\boldsymbol{x}_0).
$$

□